NEWTON-POLYGON UNIVERSALITY OF THE BOREL-SINGULARITY RADIUS FOR POLYNOMIAL CONTINUED FRACTIONS

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Abstract. We isolate, as a single citable artefact, the cross-degree identity

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}$$

for the leading Borel-plane singular distance of the formal generating function $f(z) = \sum_{n\geq 0} Q_n z^n$ of a polynomial continued fraction $(1, b)$ of degree d with leading coefficient $\beta_d > 0$. The identity is currently distributed across three SIARC artefacts: the canonical $d = 2$ Newton-polygon proof (Channel Theory v1.3, [1]), the $d = 4$ verification at dps = 80 across 8 quartic representatives with spread 0 (PCF2-SESSION-Q1, [3]), and the original $d = 2$ baseline (PCF-1 v1.3, [2]). We frame the resulting picture in three explicit epistemic tiers: (i) *proven* at $d = 2$ by the Newton-polygon argument recalled here; (ii) *verified* (in the empirical sense: a measurement at dps = 80 with spread 0 across 8 representatives) at $d = 4$; (iii) *conjectured* for general $d \geq 2$ (Conjecture 3.3.A* of [1]), with the $d = 3$ direct Newton-polygon test deferred to the operator queue. The note is purely consolidatory; no new numerical pipeline is run.

1. Setup

Throughout this note $(1, b)$ denotes a polynomial continued fraction with constant numerator $a_n \equiv 1$ and polynomial denominator

$$b(n) \;=\; \beta_d\, n^d + \beta_{d-1}\, n^{d-1} + \cdots + \beta_1\, n + \beta_0, \qquad \beta_d > 0,$$

with $d \geq 2$. Let Q_n denote the partial denominators of the PCF, satisfying the three-term recurrence $Q_n = b(n)\, Q_{n-1} + Q_{n-2}$ with $Q_{-1} = 0$, $Q_0 = 1$. The formal generating function

$$f(z) \;=\; \sum_{n\geq 0} Q_n\, z^n$$

satisfies the order-2 inhomogeneous linear ODE

$$L_d f \;:=\; \big(1 \,-\, z\, B_d(\theta + 1) \,-\, z^2\big)\, f \;=\; 1, \qquad \theta := z\,\partial_z, \tag{1}$$

with B_d the d-th degree polynomial encoding b on the shift-rescaled Euler operator θ. Equation (1) has a regular point at $z = \infty$ and an irregular singular point at $z = 0$; the Borel-plane singular distance ξ_0 is the modulus of the leading characteristic root at $z = 0$ along the slope-$1/d$ edge of the Newton polygon of the homogeneous part of (1). The scope of this note is fixed by $\beta_d > 0$ and $d \geq 2$; the $\beta_d < 0$ branch and the so-called "$d = 2$ anomaly" Galois bin ([2, Remark 5.E]) are out of scope and are recorded as open in Section 6.

2. The proven case $d = 2$

The $d = 2$ Newton-polygon argument is in canonical form in [1, Proposition 3.3.A and Worked Example §3.3], with the same content recorded earlier in [2, Theorem 5, §5]. We follow the Channel Theory v1.3 idiom because it carries the slope and characteristic-polynomial framing needed for the cross-degree extension in Sections 3 and 4.

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Proposition 2.1 (Universality of ξ_0, degree 2; [1, Prop. 3.3.A]). *For every degree-2 PCF $(1, b)$ in scope with $b(n) = \beta_2 n^2 + \beta_1 n + \beta_0$ and $\beta_2 > 0$, the leading Borel-plane singularity of the formal generating function $f(z) = \sum_{n\geq 0} Q_n z^n$ at $z = 0$ is located at distance*

$$\xi_0 \;=\; \frac{2}{\sqrt{\beta_2}}$$

from the origin, and the secondary indicial exponent of the formal solution at the irregular singular point $u = \sqrt{z} = 0$ is

$$\rho \;=\; -\frac{3}{2} \;-\; \frac{\beta_1}{\beta_2}.$$

Sketch (after [1, §3.3]*).* The Newton polygon of the homogeneous part of (1) at $z = 0$, with $d = 2$, is read off from the lattice points $\{(0, 0), (1, 0), (1, 1), (1, 2), (2, 0)\}$ (weight = order of θ, height = order of z). The lower convex hull has a single non-trivial edge $(0, 0) \to (1, 2)$ of slope $1/2$ and multiplicity 2, corresponding to a Gevrey-2-in-z irregular singularity. Substituting $z = u^2$ and $\theta = (u/2)\,\partial_u$, the operator becomes Gevrey-1-in-u, and the level-$1/u$ trans-series ansatz

$$f_\pm(u) \;=\; \exp\!\Big(\frac{c}{u}\Big)\, u^\rho \,\Big(1 + \sum_{k\geq 1} a_k u^k\Big)$$

produces the characteristic polynomial

$$\chi(c) \;=\; 1 \;-\; \frac{\beta_2}{4}\, c^2$$

at leading order, whose two roots are $c = \pm 2/\sqrt{\beta_2}$. The positive root pins $\xi_0 = 2/\sqrt{\beta_2}$. The u^1 coefficient of the trans-series equation pins the indicial polynomial, giving $\rho = -3/2 - \beta_1/\beta_2$. The Birkhoff recursion for the a_k then proceeds row-by-row in the u^k coefficient. The full computation, including the worked V_{quad} example and the (α_1, α_0)-independence of ξ_0, is [1, §3.3 Worked Example]. □

Remark 2.2 ((α_1, α_0)-independence at $d = 2$). The numerator polynomial a enters the operator L_2 only at the z^2 Newton-polygon vertex $(2, 0)$, i.e. at order u^4 in the u-uniformised operator. Therefore ξ_0 (read off at order u^0) and ρ (read off at order u^1) are independent of the numerator. The first formal invariant that does depend on α_1, α_0 is a_2, which is the splitting witness in [1, Prop. 3.3.B] for the QL01/QL02 family pair (identical $\xi_0 = 2$ and $\rho = -5/2$, disagreeing a_2).

3. The verified case $d = 4$

At $d = 4$, the same Newton-polygon construction predicts a single slope-$1/4$ edge of multiplicity 2 with characteristic polynomial $1 - (\beta_4/4^4)\, c^4$, hence the prediction $\xi_0 = 4/\beta_4^{1/4}$. This prediction has been tested computationally in [3] and is consistent with the data described below.

Proposition 3.1 (Empirical record at $d = 4$; [1, Prop. 3.3.A′], [3]). *For a degree-4 PCF $(1, b)$ with $b(n) = \beta_4 n^4 + \text{l.o.t.}$ and $\beta_4 > 0$, the Newton polygon of (1) at $z = 0$ has lattice points $\{(0, 0), (1, 0), (1, 1), (1, 2), (1, 3), (1, 4), (2, 0)\}$, with lower-left convex hull supporting the edge $E : (0, 0) \to (1, 4)$ of slope $1/4$. The WKB ansatz $f \sim \exp(c/u)$ with $z = u^4$ then pins the characteristic equation*

$$\chi(c) \;=\; 1 \;-\; \frac{\beta_4}{4^4}\, c^4 \;=\; 0,$$

whose positive real root is $c = 4/\beta_4^{1/4}$. The identity $\xi_0(b) = 4/\beta_4^{1/4}$ is consistent with the PCF2-SESSION-Q1 measurement: across 8 quartic representatives drawn from the four trichotomy bins of the session (S_4 generic, D_4 Eisenstein anchor, V_4 biquadratic anchor, plus $\beta_4 = 7$ sample), the inferred $c(4)$ has spread 0 at dps $= 80$.

The detailed Newton-polygon construction and the per-family table are in [3, `newton_polygon_d4.tex`]; the PCF2-SESSION-Q1 `claims.jsonl` entry `Q1-B` carries the SHA-256 output hash `1ad90f60...d5b10bc7` of the underlying script `session_q1_newton.py` and is the canonical AEAL anchor for the $d = 4$ measurement.

Remark 3.2 (Falsification of the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$; [1, Remark 3.3.E]). Version 1.1 of the Channel Theory outline conjectured the candidate $c(d) = 2\sqrt{(d-1)!}$ for the universality constant in $\xi_0(b) = c(d)/\beta_d^{1/d}$. That candidate matches the proven $d = 2$ value ($c(2) = 2\sqrt{1!} = 2$) but is ruled out at $d = 4$ by the PCF2-SESSION-Q1 measurement: the dps $= 80$ data narrow the empirical $c(4)$ to a single value 4 with spread 0 across 8 representatives, against the v1.1 prediction $c(4) = 2\sqrt{3!} = 2\sqrt{6} \approx 4.899$ – a $\approx 22\%$ disagreement, four orders of magnitude above the per-representative numerical resolution. The linear form $c(d) = d$ of Theorem 4.1 is the form consistent with both the proven $d = 2$ value and the $d = 4$ measurement.

4. The conjectured general-d case

Conjecture 4.1 (Cross-degree universality of ξ_0; [1, Conj. 3.3.A*])**.** *For every PCF $(1, b)$ in scope with $\deg b = d \geq 2$ and $b(n) = \beta_d n^d + \text{l.o.t.}$ ($\beta_d > 0$), the leading characteristic root of the irregular singularity at $z = 0$ of the operator L_d in (1) is*

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}.$$

Status. Proven at $d = 2$ (Theorem 2.1). *Verified* at $d = 4$ at dps $= 80$ with spread 0 across 8 representatives (Theorem 3.1). Pending direct Newton-polygon test at $d = 3$ (op:xi0-d3-direct; see Section 6). *Conjectured* for general $d \geq 2$.

Heuristic structural picture. The structural Newton-polygon picture that supports the conjecture is as follows; we record it as a heuristic, not as a proof. For any monic polynomial b of degree d, the operator L_d in (1) has Newton polygon at $z = 0$ supported on the lattice points

$$\{(0,0)\} \;\cup\; \{(1,k) : 0 \leq k \leq d\} \;\cup\; \{(2,0)\},$$

and the lower-left convex hull has a single non-trivial edge $E : (0,0) \to (1,d)$ of slope $1/d$ and multiplicity 2. Substituting $z = u^d$ with $\theta = (u/d)\,\partial_u$ and applying the trans-series ansatz $f \sim \exp(c/u)$ pins the characteristic polynomial along E to

$$\chi(c) \;=\; 1 \;-\; \frac{\beta_d}{d^d}\, c^d$$

at leading order, whose unique positive real root is $c = d/\beta_d^{1/d}$. This is the structural reason why $c(d) = d$ is the form consistent with both the proven $d = 2$ value and the $d = 4$ measurement, and why the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$ is ruled out (Theorem 3.2).

The picture above does *not* discharge the general-d case: the indicial-polynomial analysis fixing the secondary exponent ρ_d, and the Birkhoff recursion delivering the formal coefficients a_k at $d \geq 3$, are not written out here and are open even at $d = 3$ (see Section 6, item (a)).

5. Status of $d = 3$

The $d = 3$ direct Newton-polygon test is the next discrete step after the $d = 2$ proof and the $d = 4$ measurement. It is recorded as an operator queue entry in [1, §9, op:xi0-d3-direct] and is currently *deferred*, not run. The recipe is one Newton-polygon construction per Galois bin on a cubic representative $b(n) = \beta_3 n^3 + \ldots$, with the prediction $\xi_0 = 3/\beta_3^{1/3}$ to be checked at dps $= 80$ against the leading characteristic root of the slope-1/3 edge of (1); the current tractability estimate is 1–2 hours of computation [1, §9] with no halt condition expected.

A successful $d = 3$ measurement would narrow the interval of candidates for $c(d)$ further but would not by itself discharge the general-d case (Theorem 4.1); a structural proof of the indicial / Birkhoff layer at $d \geq 3$ remains open.

6. Open questions

(a) op:xi0-d3-direct ([1, §9]): direct Newton-polygon test of $\xi_0(b) = 3/\beta_3^{1/3}$ on at least one cubic representative per Galois bin, at dps = 80. Predicted runtime: 1–2 hours per bin. No halt condition is expected; a measurement consistent with the prediction would narrow the empirical interval for $c(d)$ to $\{2, 3, 4\}$ at $d \in \{2, 3, 4\}$ and would support Theorem 4.1 along the linear arithmetic progression. A measurement inconsistent with the prediction would falsify the conjecture and trigger the corresponding SIARC halt condition.

(b) The $d \geq 5$ case. No empirical record exists for $d \geq 5$; the Newton-polygon construction (Section 4) extends in principle, but neither the slope-$1/d$ edge analysis at finite β_d nor the secondary indicial exponent ρ_d has been written out. A combinatorial generalisation of Theorem 2.1 to all $d \geq 2$ – including a uniform proof of the $(\alpha_1, \ldots, \alpha_{d-2})$-independence of ξ_0 (Theorem 2.2) – is the natural next target.

(c) The $\beta_d < 0$ branch and the $d = 2$ anomaly Galois bin ([2, Remark 5.E]). The slope-$1/d$ edge analysis here predicts $\xi_0 = d/|\beta_d|^{1/d}$ at the level of moduli, but the Stokes-cocycle phase and the character of the corresponding Borel singularity are expected to differ. A separate analysis is open.

AI Disclosure

The SIARC v1.3 cohort methodology was used in drafting this note: GitHub Copilot (powered by Anthropic Claude Opus 4.7) and Anthropic Claude were used for prose drafting and consistency-checking; all theorem statements, conjecture formulations, and editorial decisions remain the author's, and every numerical claim traces to an AEAL entry in `claims.jsonl` with a SHA-256 output hash on the SIARC bridge under `sessions/2026-05-02/D2-NOTE-DRAFT/`.

References

[1] Mauricio Echizen Kubo. Channel theory for polynomial continued fractions, v1.3: Asymptotic channels, the $\xi_0 = 2/\sqrt{\beta_2}$ identity, and a bridge conjecture, 2026. Concept DOI 10.5281/zenodo.19941678. Carries the canonical Newton-polygon proof of $\xi_0 = 2/\sqrt{\beta_2}$ at $d = 2$ (Prop. 3.3.A), the empirical record of $\xi_0 = 4/\beta_4^{1/4}$ at $d = 4$ (Prop. 3.3.A′), the cross-degree universality conjecture (Conj. 3.3.A*), the falsification of the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$ at $d = 4$ (Remark 3.3.E), and the $d = 3$ deferred operator entry op:xi0-d3-direct (§9). PDF SHA-256 df3b90e808e49e84fbba53e5663a851256303496fc1536fefbf962aba2ebdc18.

[2] Papá-Nokechi. PCF-1 v1.3: Stokes signal and channel scope for $\delta < 0$ degree-2 polynomial continued fractions, 2026.

[3] Papá-Nokechi. PCF-2 Session Q1: quartic harvest at $d = 4$ and the universal newton-polygon characteristic root $\xi_0(b) = d/\beta_d^{1/d}$. SIARC bridge, sessions/2026-05-01/PCF2-SESSION-Q1/, 2026.

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